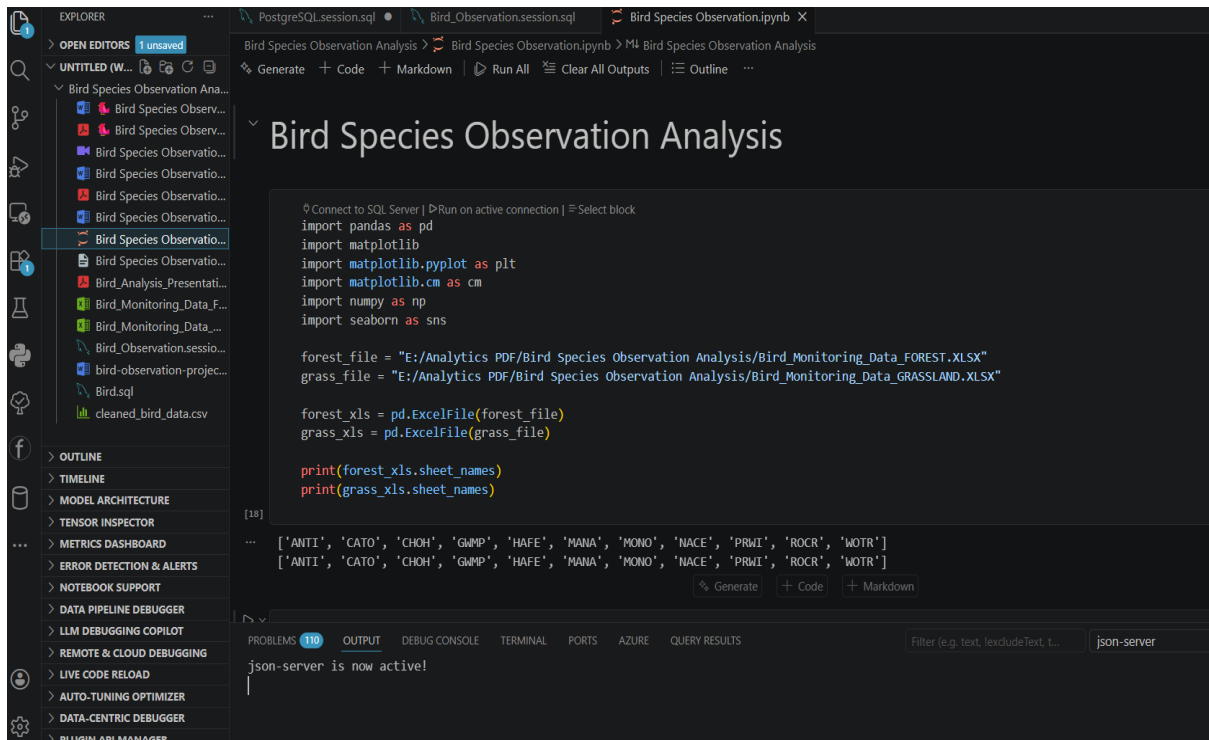


Bird Species Observation Screenshots

Python:



This screenshot shows the first part of a Python script in VS Code. The Explorer panel on the left shows a project named 'Bird Species Observation Analysis' with several files. The main editor displays the following code:

```
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.cm as cm
import numpy as np
import seaborn as sns

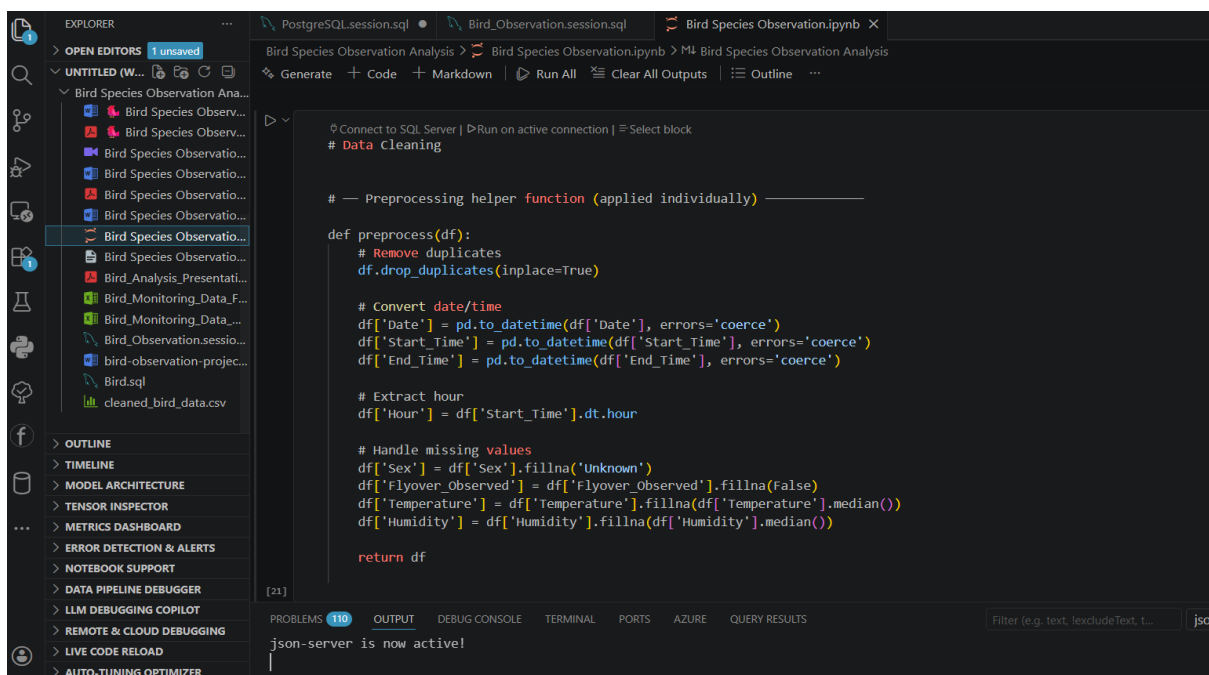
forest_file = "E:/Analytics PDF/Bird Species Observation Analysis/Bird_Monitoring_Data_FOREST.XLSX"
grass_file = "E:/Analytics PDF/Bird Species Observation Analysis/Bird_Monitoring_Data_GRASSLAND.XLSX"

forest_xls = pd.ExcelFile(forest_file)
grass_xls = pd.ExcelFile(grass_file)

print(forest_xls.sheet_names)
print(grass_xls.sheet_names)
```

The output panel at the bottom shows the results of the print statements:

```
[18]
...
['ANTI', 'CATO', 'CHOH', 'GWMP', 'HAFE', 'MANA', 'MONO', 'NACE', 'PRWI', 'ROCR', 'WOTR']
['ANTI', 'CATO', 'CHOH', 'GWMP', 'HAFE', 'MANA', 'MONO', 'NACE', 'PRWI', 'ROCR', 'WOTR']
```



This screenshot shows the second part of the Python script in VS Code. The Explorer panel on the left shows the same project. The main editor displays the following code:

```
# Connect to SQL Server | Run on active connection | Select block
# Data cleaning

# — Preprocessing helper function (applied individually) —————

def preprocess(df):
    # Remove duplicates
    df.drop_duplicates(inplace=True)

    # Convert date/time
    df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
    df['Start_Time'] = pd.to_datetime(df['Start_Time'], errors='coerce')
    df['End_Time'] = pd.to_datetime(df['End_Time'], errors='coerce')

    # Extract hour
    df['Hour'] = df['Start_Time'].dt.hour

    # Handle missing values
    df['Sex'] = df['Sex'].fillna('Unknown')
    df['Flyover_Observed'] = df['Flyover_Observed'].fillna(False)
    df['Temperature'] = df['Temperature'].fillna(df['Temperature'].median())
    df['Humidity'] = df['Humidity'].fillna(df['Humidity'].median())

    return df
```

The output panel at the bottom shows the results of the print statements:

```
[21]
...
json-server is now active!
```

EXPLORER

OPEN EDITORS 1 unsaved

UNTITLED (W...)

Bird Species Observation Ana...

Bird Species Observ...

Bird Species Observ...

Bird Species Observ...

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Bird Species Observ...

Bird Analysis Presentati...

Bird_Monitoring_Data_F...

Bird_Monitoring_Data_...

Bird_Observation.sessio...

bird-observation-projec...

Bird.sql

cleaned_bird_data.csv

OUTLINE

TIMELINE

MODEL ARCHITECTURE

TENSOR INSPECTOR

METRICS DASHBOARD

ERROR DETECTION & ALERTS

NOTEBOOK SUPPORT

DATA PIPELINE DEBUGGER

LLM DEBUGGING COPILOT

REMOTE & CLOUD DEBUGGING

LIVE CODE RELOAD

AUTO-TUNING OPTIMIZER

DATA-CENTRIC DEBUGGER

PostgreSQL.session.sql

Bird_Observation.session.sql

Bird Species Observation.ipynb

Bird Species Observation Analysis

Generate + Code + Markdown Run All Clear All Outputs Outline

EDA (Exploratory Data Analysis)

Connect to SQL Server | Run on active connection | Select block

Birds Count by Location

```
print(df.groupby('Admin_Unit')['Common_Name'].count().sort_values(ascending=False))
```

```
location_counts = df.groupby('Admin_Unit')['Common_Name'].count()
```

```
plt.figure()
location_counts.plot(kind='bar')
plt.title("Bird Counts by Location")
plt.xlabel("Location")
plt.ylabel("Count")
plt.show()
```

[35]

Admin_Unit	Count
ANTI	3463
MONO	2522
PRWI	2463
CHOH	2202
MANA	1901
CATO	805
NACE	684
...	...

PROBLEMS 110 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE QUERY RESULTS

Filter (e.g. text, excludeText, t...) json-se

json-server is now active!

EXPLORER

OPEN EDITORS 1 unsaved

UNTITLED (W...)

Bird Species Observation Ana...

Bird Species Observ...

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Bird Species Observ...

Bird Analysis Presentati...

Bird_Monitoring_Data_F...

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Bird Species Observation.ipynb

Bird Species Observation Analysis

Generate + Code + Markdown Run All Clear All Outputs Outline

WOTR 127

Name: Common_Name, dtype: int64

...

Bird Counts by Location

Location	Count
ANTI	3463
CATO	805
CHOH	2202
GWMP	400
HAFE	500
MANA	1901
MONO	2522
NACE	684
PRWI	2463
ROCR	300
WOTR	100

PROBLEMS 110 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE QUERY RESULTS

Filter (e.g. text, excludeText, t...) json

json-server is now active!

EXPLORER

OPEN EDITORS 1 unsaved

UNTITLED (W...)

Bird Species Observation Ana...

Bird Species Observ...

Bird Species Observ...

Bird Species Observ...

Bird Species Observ...

Bird Species Observ...

Bird Species Observ...

Bird Species Observ...

Bird Species Observ...

Bird Species Observ...

Bird Analysis Presentati...

Bird_Monitoring_Data_F...

Bird_Monitoring_Data_...

Bird_Observation.sessio...

bird-observation-projec...

Bird.sql

cleaned_bird_data.csv

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Bird Species Observation.ipynb

Bird Species Observation Analysis

Generate + Code + Markdown Run All Clear All Outputs Outline

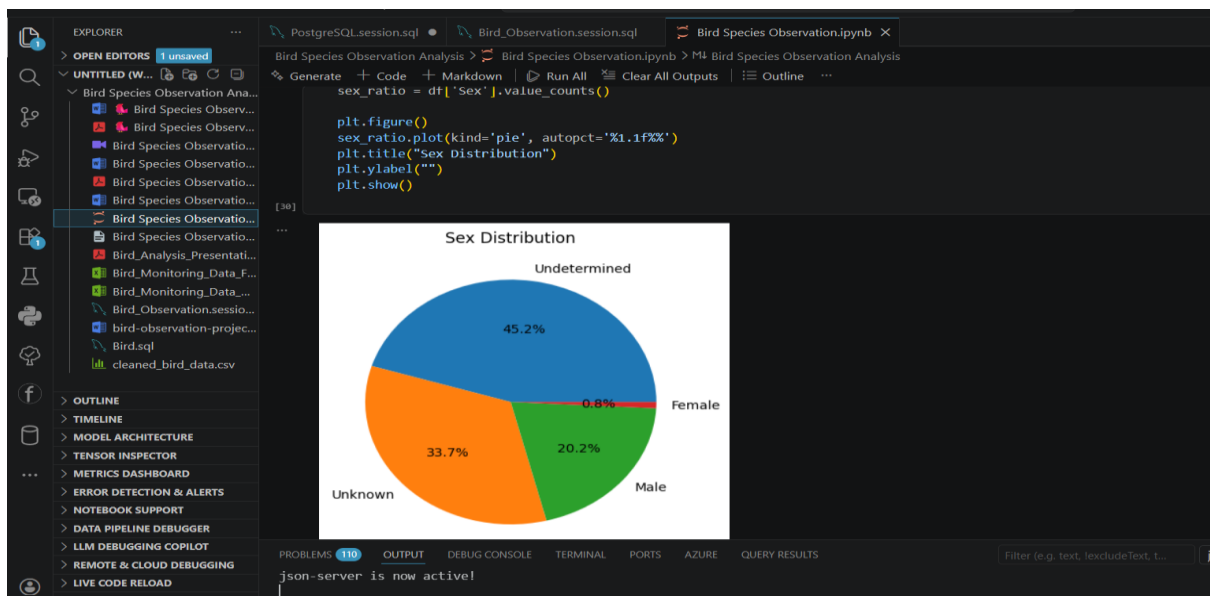
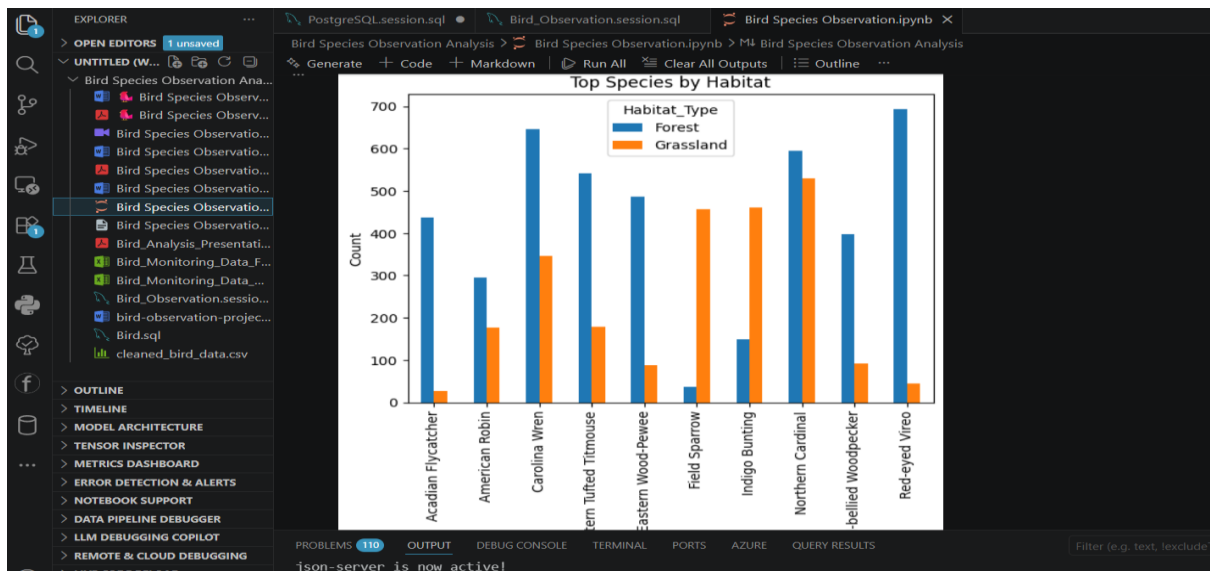
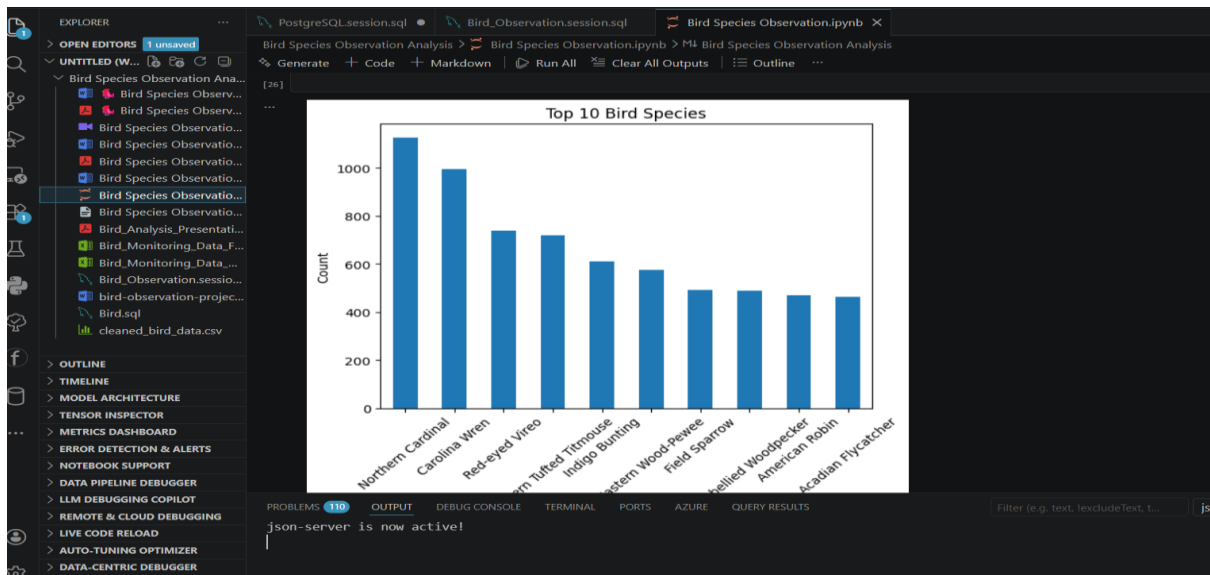
Forest vs Grassland Bird Count

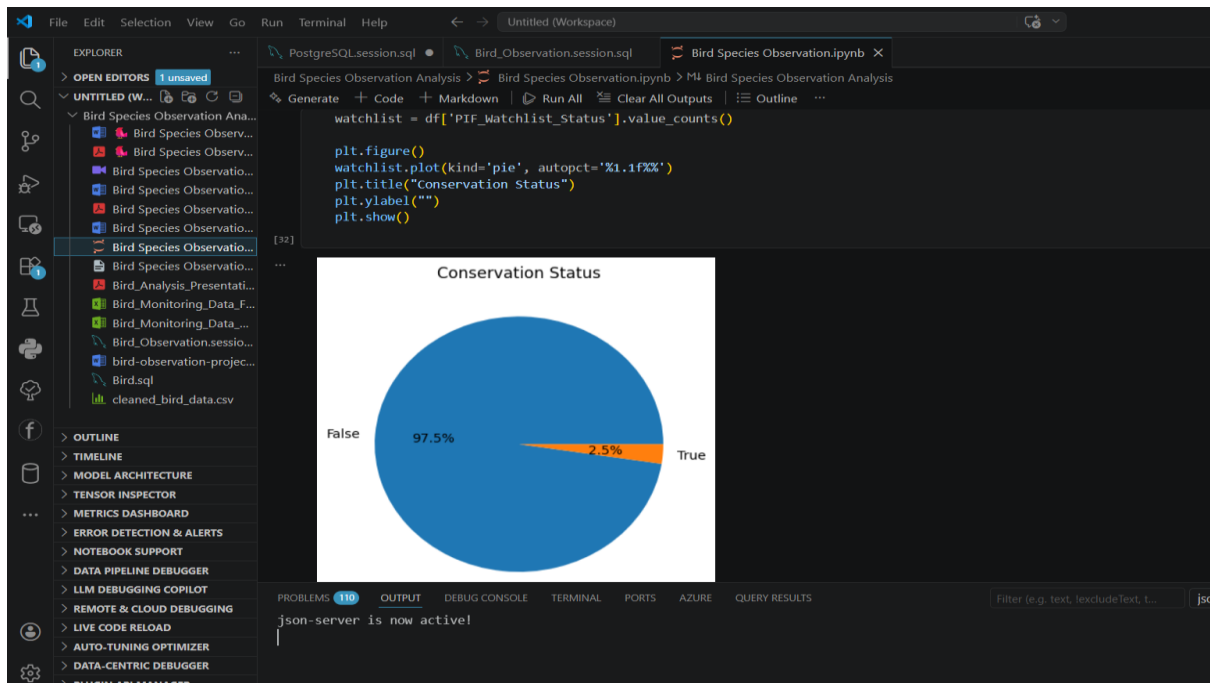
Habitat Type	Number of Birds
Forest	8000
Grassland	6800

PROBLEMS 110 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE QUERY RESULTS

Filter (e.g. text, excludeText, t...) json-ser

json-server is now active!





Postgres SQL

The screenshot shows a PostgreSQL session in VS Code. The SQL queries and their results are as follows:

```
45 copy bird_data
46 FROM 'E:/Analytics PDF/Bird Species Observation Analysis/cleaned_bird_data.csv'
47 DELIMITER ','
48 CSV HEADER
49 QUOTE '''
50 ESCAPE '\';
51
52 SELECT COUNT(*) FROM bird_data;
53
54 SELECT * FROM bird_data LIMIT 10;
55
56 ALTER TABLE bird_data
57 ALTER COLUMN Temperature TYPE FLOAT USING Temperature::FLOAT;
58
59 -- Business Queries Analysis
60
61 -- Forest vs Grassland comparison: Which habitat has more birds?
62
63
64 SELECT Habitat_Type, COUNT(*) AS total_birds
65 FROM bird_data
66 GROUP BY Habitat_Type
67 ORDER BY total_birds DESC;
68
69 -- Biodiversity comparison: Which habitat has more species diversity?
70 SELECT Habitat_Type, COUNT(DISTINCT Scientific_Name) AS species_count
71 FROM bird_data
72 GROUP BY Habitat_Type
73 ORDER BY species_count DESC;
74
```

The results of the queries are shown in the right-hand pane:

habitat_type	total_birds
Forest	8546
Grassland	6826

SQLTOOLS ... PostgreSQLsession.sql • Bird_Observation.session.sql X

File Edit Selection View Go Run Terminal Help < -> Untitled (Workspace)

CONNECTIONS

- Bird_Observation p...
- bird_observation data...
- Schemas
- public
- Tables
- bird_data
 - admin_unit_co...
 - sub_unit_code...
 - site_name TEXT
 - plot_name TEXT
 - location_type...
 - year INTEGER
 - date TIMESTA...
 - start_time TIM...
 - end_time TIME...
 - observer TEXT
 - visit INTEGER
 - interval_length...
 - id_method TEXT

BOOKMARKS

QUERY HISTORY

- Bird_Observation
 - SELECT Common_Name, ...
 - SELECT Temperature, CO...
 - SELECT Habitat_Type, CO...

Bird Species Observation Analysis > Bird_Observation.session.sql

```
132 FROM bird_data
133 GROUP BY Month
134 ORDER BY Month;
135
136 -- Yearly trend
137 SELECT Year, COUNT(*) AS total_birds
138 FROM bird_data
139 GROUP BY Year
140 ORDER BY Year;
141
142 -- SPECIES ANALYSIS:
143
144 -- Total unique species
145 SELECT COUNT(DISTINCT Scientific_Name) AS total_species
146 FROM bird_data;
147
148 -- Species richness per location
149 SELECT Admin_Unit, COUNT(DISTINCT Scientific_Name) AS species_col
150 FROM bird_data
151 GROUP BY Admin_Unit
152 ORDER BY species_count DESC;
153
154 -- Species distribution (Top 10)
155 SELECT Common_Name, COUNT(*) AS count
156 FROM bird_data
157 GROUP BY Common_Name
158 ORDER BY count DESC
159 LIMIT 10;
160
161 -- Sex ratio
```

common_name count

common_name	count
Northern Cardinal	1125
Carolina Wren	993
Red-eyed Vireo	738
Eastern Tufted Titmouse	720
Indigo Bunting	611
Eastern Wood-Pewee	574
Field Sparrow	492
Red-bellied Woodpecker	489
American Robin	470
Acadian Flycatcher	462

CONSOLE RE-RUN QUERY EXPORT OPEN

PROBLEMS 110 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE QUERY RESULTS SQL CONSOLE

SELECT successfully executed. 11:21:50 PM

SQLTOOLS ... PostgreSQLsession.sql • Bird_Observation.session.sql X

File Edit Selection View Go Run Terminal Help < -> Untitled (Workspace)

CONNECTIONS

- Bird_Observation p...
- bird_observation data...
- Schemas
- public
- Tables
- bird_data
 - admin_unit_co...
 - sub_unit_code...
 - site_name TEXT
 - plot_name TEXT
 - location_type...
 - year INTEGER
 - date TIMESTA...
 - start_time TIM...
 - end_time TIME...
 - observer TEXT
 - visit INTEGER
 - interval_length...
 - id_method TEXT

BOOKMARKS

QUERY HISTORY

- Bird_Observation
 - SELECT Temperature, CO...
 - SELECT Habitat_Type, CO...
 - CREATE TABLE bird_data/

Bird Species Observation Analysis > Bird_Observation.session.sql

```
79 ORDER BY total_birds DESC;
80
81 -- Dominant species: Which species are most common?
82 SELECT Common_Name, COUNT(*) AS count
83 FROM bird_data
84 GROUP BY Common_Name
85 ORDER BY count DESC
86 LIMIT 10;
87
88 -- Habitat preference: Which species prefer which habitat?
89 SELECT Habitat_Type, Common_Name, COUNT(*) AS count
90 FROM bird_data
91 GROUP BY Habitat_Type, Common_Name
92 ORDER BY Habitat_Type, count DESC;
93
94 -- WEATHER IMPACT :
95
96 --How does temperature affect bird activity?
97 SELECT Temperature, COUNT(*) AS bird_count
98 FROM bird_data
99 GROUP BY Temperature
100 ORDER BY Temperature;
101
102 -- How does humidity affect bird activity?
103 SELECT Humidity, COUNT(*) AS bird_count
104 FROM bird_data
105 GROUP BY Humidity
106 ORDER BY Humidity;
107
108 -- Weather conditions vs activity
```

temperature bird_count

temperature	bird_count
11	20
11.1999998092651	6
11.39999996185303	12
11.6000003814697	3
11.8000001907349	6
12	15
12.89999996185303	28
13	12
13.1999998092651	9
13.8000001907349	39
13.89999996185303	8
14	90
14.1000003814697	13
14.3000001907349	41
14.39999996185303	26
14.5	36
14.6000003814697	13
14.6999998092651	38
14.8000001907349	47

CONSOLE RE-RUN QUERY EXPORT OPEN

PROBLEMS 110 OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE QUERY RESULTS SQL CONSOLE

SELECT successfully executed. 11:21:08 PM

POWER BI

